VTVL Hopper Engineering Notebook

Contents

1 Design Requirements

1.1 Mission Objective

Design a throttleable, simple liquid bipropellant engine (and every subsystem required for it to work) for a minimal VTVL hopper. The engine exists to support a possible vehicle in the future including subsystems such as avionics, landing gear, and test instrumentation. The theoretical flight objective for the full vehicle to be designed later is controlled liftoff, hover or near-hover control authority, and safe landing.

1.2 Requirement Classification

The requirements below are limited to items that must be true before the design is considered acceptable. Quantities that depend on vehicle mass convergence, tank selection, injector pressure drop, cooling margin, or throttle hardware are carried as sizing variables, not fixed requirements.

Item	Classification	Requirement / design rule
Budget	Hard	Prototype propulsion hardware target: ≤ $5,000 excluding propellant
Architecture	Hard	No turbopumps.
Cryogenics	Hard	No cryogenic propellant storage.
Hazard class and Legality	Hard	Nothing specifically potentially hazardous, corrosive, toxic, or illegal.
Material access	Hard	Primary wetted structural materials shall be readily obtainable aluminum alloy or stainless steel unless a later section gives a specific compatibility basis.
Fluid connectors	Hard	Use pressure-rated commercial connector families only: AN/MS, CGA, Swagelok-compatible compression fittings, or equivalent documented hardware.
Pressure safety	Hard	Pressure-wetted custom parts shall use burst factor of safety ≥ 4 against MEOP. Purchased cylinders, valves, regulators, and fittings shall be used only within documented manufacturer ratings.
Remote operation	Hard	All oxidizer and hot-fire operations shall be controllable from a safe remote location with positive propellant isolation before personnel approach.
Throttle demonstration	Test requirement	The engine shall demonstrate repeatable operation at full thrust and at lower commanded setpoints before flight. The final throttle ratio is set after vehicle mass is known.
Combustion stability	Test requirement	No sustained hard-start, chugging, buzz, or high-frequency instability is acceptable in qualification hot-fire data. Stability criteria are finalized in the injector and test sections.
TVC interface	Layout requirement	Reserve a defined thrust-frame or gimbal interface. Cooling jacket and plumbing geometry shall not block later TVC integration.

Table 1: Design requirements for the first prototype phase.

Quantity	Status	How it is used in this notebook
Nominal thrust	Sizing variable	Preliminary analyses may bracket 200-400 N, but the final value belongs in the engine sizing section after dry mass and propellant mass converge.
Full-thrust liftoff margin	Derived later	After m_0 is known, require $\frac{T_{\max}}{m_0 g_0} \geq 1.2$ minimum. Higher margin is useful only if it does not force unnecessary system scale-up.
Minimum throttle	Derived later	After m_0 is known, require $T_{\min} < m_0 g_0$ so the vehicle can command descent.
Burn duration	Design goal	30 s minimum useful demonstration; 60 s stretch target. Propellant mass is computed after thrust and $I_{\mathrm{sp,SL}}$ are selected.
Chamber pressure	Design variable	Use 150 psia as the first static-fire analysis point and carry 200 psia as a performance-upgrade case if feed, cooling, and structure close.

Table 2: Sizing variables intentionally deferred to later notebook sections.

2 Propellant

2.1 Define the Problem

Select a storable oxidizer/fuel pair for a small pressure-fed VTVL hopper. The selected pair must support repeatable throttling, safe ground testing, affordable consumables, realistic tankage, and a credible path to injector and cooling design. The decision is not "maximize I_{sp}"; it is "minimize total development risk while preserving enough performance for a minimal controlled hop."

Categories of Analysis

- Cost
- Storage mass and volume
- Stability/Predictability
- Development risk
- Performance
- Complexity

2.2 Brainstorm Solutions

2.2.1 Hard Screens

These families fail a fixed constraint and are removed before detailed trade scoring.

Candidate	Reason for removal
LOX-based combinations	They violate the no-cryogenic first-article constraint. A LOX engine is a valid later project, not the fastest safe path to this hopper. Reconsidered quantitatively in Section 2.8; the screen stands.
Hydrazine, MMH, UDMH, NTO	The toxicity and legal burden are incompatible with the intended small-team, low-budget test program.
RFNA/alcohol	Corrosive oxidizer handling and compatibility burden are not acceptable for the preferred aluminum/stainless build.
90%+ H_2O_2 / alcohol	High-test peroxide is a specialized oxidizer with severe decomposition and contamination hazards; sourcing and handling dominate the project.

2.2.2 Feasible Solutions

Candidate family	Why it is worth considering	Initial concerns to test
N_2O/alcohols	Self-pressurizing oxidizer, compact liquid storage, non-cryogenic, active small-VTVL precedent.	Two-phase injector flow, tank-temperature sensitivity, oxidizer pressure decay during burn.

GOX/alcohols	Single-phase oxidizer metering, stable regulator-fed pressure, easier analytical injector sizing.	Large high-pressure gas volume, regulator cost, oxygen cleaning, tank mass.
H_2O_2/alcohols	Dense storable oxidizer; 85% peroxide performance is close to N_2O/alcohol in the sea-level ambient comparison.	High-concentration peroxide sourcing, catalyst/ignition hardware, decomposition compatibility, hazardous transport.
N_2O or GOX with methanol	Methanol is cheap and historically common in rocket fuels.	Toxicity and no computed performance advantage over IPA/ethanol.
N_2O with gaseous fuels	Some spacecraft and hybrid literature exists; gaseous fuels can simplify atomization.	Poor fuel storage density and pressure-vessel mass for a 60 s ground hopper.
N_2O/RP-1 or kerosene	Dense fuel; common hydrocarbon fuel.	Coking/soot and regenerative cooling complexity are poor matches for a small reusable engine.

2.3 Screen the Candidate Set

The table below is generated from the companion RocketCEA project at $P_c = 150$ psia, $P_{\text{amb}} = 14.7$ psia, and a sea-level-matched nozzle. Values are shifting-equilibrium RocketCEA/NASA CEA outputs [1].

Candidate	$I_{\text{sp,SL}}$	**O/F**	ε	T_c **K**	c^* **m/s**	**Screen result**
GOX / IPA	230.1 s	1.65	2.43	3237	1762	Keep as GOX finalist. Best computed sea-level impulse, but GOX storage/hardware dominates.
GOX / Ethanol	225.6 s	1.55	2.45	3187	1727	Keep as GOX finalist. Slightly lower computed I_{sp} than GOX/IPA but stronger ignition/cooling evidence.
GOX / Methanol	221.3 s	1.20	2.46	3074	1693	Remove. No advantage over GOX/ethanol or GOX/IPA, with worse toxicity.
N_2O / IPA	207.0 s	5.10	2.37	3120	1590	Keep. Best N_2O/alcohol computed result and strongest VTVL heritage.
N_2O / Ethanol	204.8 s	4.65	2.38	3064	1573	Keep. Strong static-fire literature and better coolant conductivity.
85% H_2O_2 / IPA	203.2 s	5.65	2.37	2553	1562	Near miss only. Dense oxidizer but sourcing/catalyst/decomposition burden is high.
N_2O / Methanol	203.0 s	3.50	2.38	2972	1559	Remove. No performance advantage and worse toxicity.
85% H_2O_2 / Ethanol	201.1 s	5.00	2.36	2507	1547	Near miss only; same peroxide concerns.
70% H_2O_2 / IPA	186.9 s	7.20	2.31	2184	1441	Remove. Low performance without solving peroxide access/catalyst risk.
70% H_2O_2 / Ethanol	185.0 s	6.30	2.30	2139	1426	Remove. Low performance without solving peroxide access/catalyst risk.

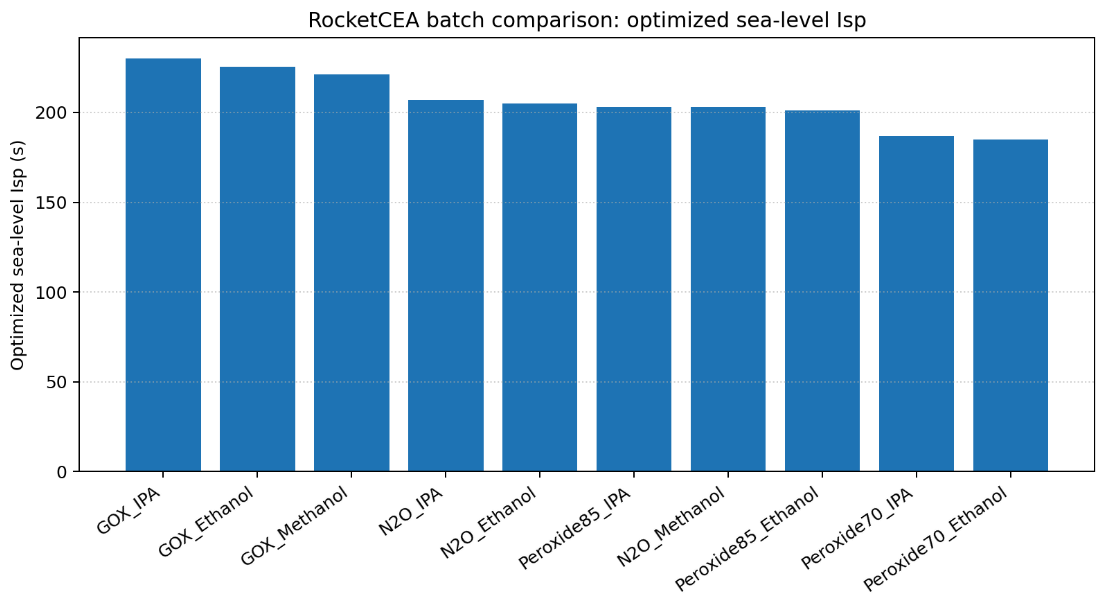

Figure 1: Generated RocketCEA comparison for optimized sea-level impulse at $P_c = 150$ psia, $P_{\text{amb}} = 14.7$ psia, shifting-equilibrium chemistry, and $P_e \approx P_{\text{amb}}$ nozzle sizing [1].

2.4 Quantitative Sourcing and System Sizing

Pricing below is a planning model, not a purchase instruction. Industrial gas prices vary by region, account status, deposit/rental policy, hazmat shipping, and cylinder exchange rules. The useful engineering comparison is the order of magnitude: GOX consumables can be cheap, but the storage/regulator system is expensive and heavy; N_2O consumables are more expensive per kilogram, but the liquid storage is much smaller and the tank is also the pressure source. I should replace these bands with local quotes before procurement, but the bands below are anchored to published prices — including the Friends of Amateur Rocketry propellant list, which is the closest thing to a posted market price for amateur-scale rocket oxidizers and includes tax and delivery to a real test site [2].

Material	Practical acquisition route	Planning price	Design note
N_2O	Racing-gas cylinder from an amateur test site or gas supplier; motorsport refill	\$12–\$22/kg	FAR sells a 56 lb racing-grade cylinder with dip tube for \$292, about \$11.5/kg [2]; motorsport refills run \$8.49–\$9.79/lb (\$18.7–\$21.6/kg) [3]. Higher one-off retail fills are reported.
GOX	Welding oxygen cylinder exchange or industrial gas supplier	\$4–\$16/kg	Welding-trade guides put small-cylinder refills/exchanges around \$4–\$12/kg [4]; FAR sells a filled 125 cf cylinder for \$71.84 including tax and delivery, about \$15/kg [2].
LOX	Industrial gas supplier dewar; also sold at amateur test sites	\$2–\$4/kg	Screened out for the first article but priced for honesty: FAR sells 230 L (roughly 262 kg) for \$540 [2], and a trade source reports a \$200 quote for a 180 L dewar [5]. Dewar deposit/rental and boiloff excluded. See Section 2.8.
99% IPA	Hardware, janitorial, lab, or chemical supplier	\$8–\$12/kg	Representative 99% IPA is \$23.50/gal before shipping/hazmat; 70% rubbing alcohol is not acceptable [6].
Anhydrous ethanol	Industrial/lab chemical supplier; denatured 200-proof acceptable if compatible	\$13–\$30/kg	Denatured 200-proof ethanol can be near \$40–\$57/gal, while tax-paid absolute ethanol can be much higher [7], [8].

70–85% H_2O_2	Specialty chemical supplier; high-concentration grades usually quote/hazmat	$20–$60+/kg placeholder	Public lower-concentration lab peroxide pricing is visible, but 70% is specialty/quote and not ordinary pool-supply grade [9], [10], [11].

Case	Thrust	Total prop.	Oxidizer	Fuel	Ox storage	Propellant cost
N_2O / IPA	200 N	5.91 kg	4.94 kg	0.97 kg	7.9 L	$67–$120
N_2O / IPA	400 N	11.82 kg	9.88 kg	1.94 kg	15.8 L	$134–$241
N_2O / Ethanol	200 N	5.97 kg	4.92 kg	1.06 kg	7.9 L	$73–$140
N_2O / Ethanol	400 N	11.95 kg	9.83 kg	2.11 kg	15.7 L	$145–$280
GOX / IPA	200 N	5.32 kg	3.31 kg	2.01 kg	17.5 L	$29–$77
GOX / IPA	400 N	10.64 kg	6.62 kg	4.01 kg	34.9 L	$59–$154
GOX / Ethanol	200 N	5.42 kg	3.30 kg	2.13 kg	17.4 L	$41–$117
GOX / Ethanol	400 N	10.85 kg	6.59 kg	4.25 kg	34.8 L	$82–$233

The GOX volume in the table assumes storage near 2200 psia and 293 K using the ideal-gas law; the N_2O volume assumes saturated-liquid storage near room temperature with a 20% ullage/design margin. The 750 kg/m³ N_2O density is deliberately a warm-tank planning value: saturated-liquid density is roughly 745–790 kg/m³ over 20–25°C and falls steeply toward the critical point, so a hot Las Vegas tank holds less than a cool one [12]. These are comparison numbers, not final pressure-vessel dimensions; candidate pressure vessels still need manufacturer ratings, valve details, fill rules, and compatibility checks [13].

The 400 N, 60 s case is the stretch corner of the requirement, not the requirement, so it must not size the architecture by itself; the honest way is to map every corner of the thrust/burn-time envelope onto real, priced vessels. Stored GOX assumes blowdown from 2265 psia service pressure to a 500 psia regulator floor, leaving about 78% of the fill usable; the N_2O planning load is 1.25× the burned mass to cover retained vapor, pressure sag near depletion, and fill tolerance [13].

Case	GOX burned / stored	GOX vessel (price, tare)	N_2O burned / loaded	N_2O bottle (price, tare)
200 N, 30 s	1.65 / 2.12 kg	80 cf steel ($143.50)	2.47 / 3.09 kg	10 lb bottle ($314.99, 6.3 kg)
200 N, 60 s	3.30 / 4.23 kg	125 cf steel ($194.50, 26.3 kg)	4.94 / 6.18 kg	15 lb bottle ($374.99, 8.1 kg)
400 N, 30 s	3.30 / 4.23 kg	125 cf steel ($194.50, 26.3 kg)	4.94 / 6.18 kg	15 lb bottle ($374.99, 8.1 kg)
400 N, 60 s	6.59 / 8.46 kg	250 cf steel ($274.80, 49.0 kg)	9.88 / 12.36 kg	2 × 15 lb bottles ($749.98, 16.1 kg)

Cylinder and bottle picks, prices, and tares are current public listings [14], [15], [16], [17]. Read this table two ways. As ground-support hardware, GOX storage is cheap at every requirement point: the 30 s floor case is one $143.50 welding cylinder, and only the full stretch case forces the 250-class cylinder. As onboard hardware, the same column is disqualifying: the mapped steel cylinders carry roughly 26–49 kg of steel to deliver 3.3–6.6 kg of usable oxygen, about 7 kg of vessel per kilogram of oxidizer against about 1.5 kg/kg for the nitrous bottles. Aluminum and composite oxygen cylinders improve the ratio but remain far from the bottle numbers and cost more, and breathing-air composite cylinders (SCBA, paintball) are not rated for oxygen service and are not a shortcut.

The vessel is not where the architectures diverge on cost, so the next table itemizes the full oxidizer-side feed system at the 200 N, 60 s reference. Allowance rows are marked as such; everything else is a current public listing.

Item	GOX / alcohol	N_2O / alcohol
Storage vessel	125 cf steel CGA-540 cylinder: $194.50, 26.3 kg tare [14].	15 lb aluminum bottle with high-flow valve: $374.99, 8.1 kg tare [17].
Pressure regulation	High-flow high-delivery oxygen regulator. Harris 3000-2500 lists $699 [18]; Victor SR4J-540 lists $1,077, rated 16,600 scfh against roughly 5,300 scfh needed at 200 N and 10,500 scfh at 400 N [19].	None. Tank vapor pressure is the feed source.

Run valve and plumbing	Oxygen-clean actuated run valve, check valve, relief, CGA-540 pigtail: \$250–\$450 allowance (quote items).	Motorsport N_2O solenoid or actuated ball valve plus -AN plumbing: \$250–\$550 allowance (quote items).
Fill and ground handling	Cylinder exchange at the supplier; no owned fill hardware.	Fill adapter and scale: \$100–\$250 allowance, or per-fill service at a speed shop [3].
Cleanliness	Every wetted part oxygen-cleaned per CGA G-4.1: \$50–\$150 consumables and verification [20].	No separate line item, but the same discipline applies: hydrocarbon contamination catalyzes N_2O decomposition [21].
Subtotal	**\$1,190–\$1,870**	**\$725–\$1,175**

Shared subsystems are excluded from both columns because both architectures need them: fuel tank, fuel-side pressurant, commodity pressure transducers (\$30–\$80 class), harness, and test stand. One common claim needs correcting here: N_2O does not eliminate the pressurant bottle, only the oxidizer-side one. The AEL Snark vehicle fed IPA from a nitrogen-pre-pressurized tank, and this project will need the same small N_2 system either way [22].

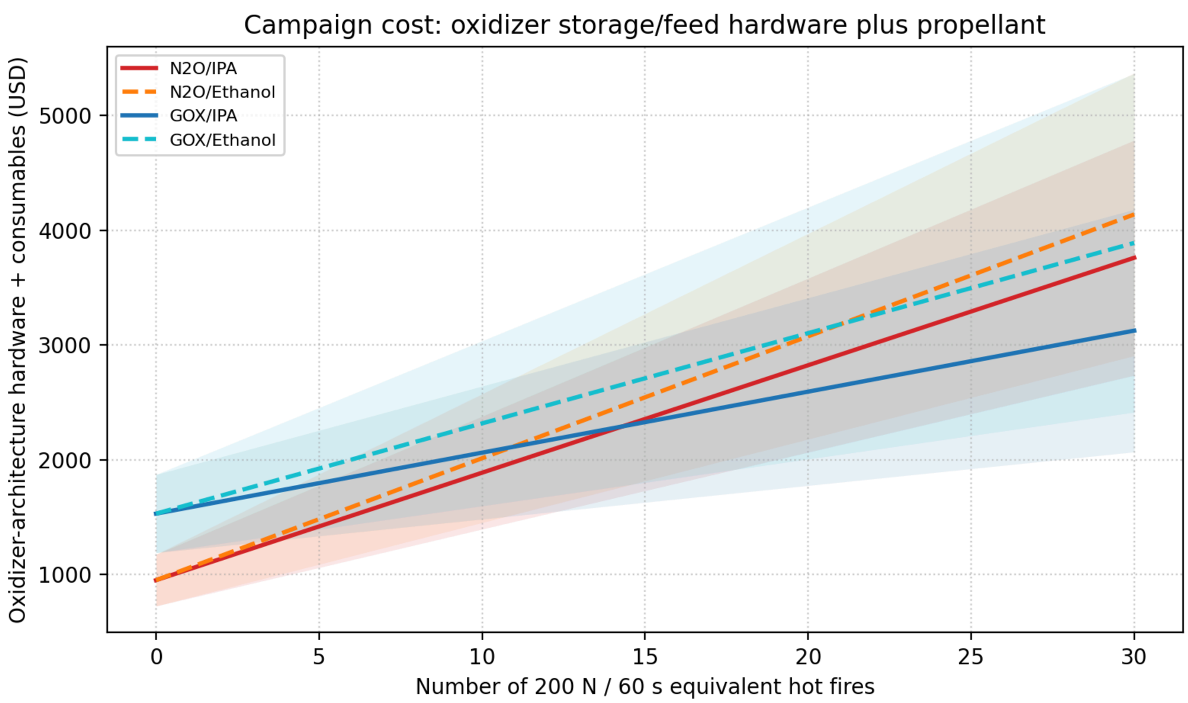

Figure 2: Oxidizer-architecture hardware plus consumables against number of 200 N, 60 s hot fires; mid-band lines with low/high shading [13].

The hardware premium and the consumables premium pull in opposite directions, so the honest comparison is the whole campaign. The GOX architecture costs roughly \$470–\$700 more up front, almost all of it regulator and oxygen-clean components rather than the cylinder. The N_2O architecture pays roughly \$38–\$43 more per test than GOX/IPA and \$4–\$26 more than GOX/ethanol, whose expensive fuel erodes most of the cheap-gas advantage. On mid-band assumptions the GOX premium pays back after about 15 hot fires against GOX/IPA and only after about 40 against GOX/ethanol. For a pure static-fire program, GOX/IPA is the cheapest campaign. What that framing misses is the flight article: the compact bottle in the mapping table above is the only storage option in this set that a hopper can plausibly lift, and it doubles as the pressurization system.

2.5 Peroxide and Methanol Near-Misses

85% H_2O_2/alcohol is not dismissed because of sea-level performance alone. In the baseline comparison it is close to N_2O/alcohol: 85% H_2O_2/IPA gives 203.2 s, only 3.8 s below N_2O/IPA [1]. It is removed because the supporting architecture is worse for this project: high-concentration peroxide sourcing is specialized, material compatibility is narrow, decomposition contamination can be violent, and practical engines often need a catalyst or dedicated ignition/decomposition hardware [11], [23].

70% H_2O_2 is not ordinary pool-supply grade in the sense needed here. Even if a 70% source is found, sea-level ambient-corrected performance is about 185–187 s at the baseline condition, while retaining the peroxide compatibility and handling burden [1].

Methanol combinations are also removed. Methanol does not outperform the matching IPA/ethanol cases in the baseline comparison, and the CDC/NIOSH pocket guide lists skin absorption as an exposure route, OSHA PEL of 200 ppm, and symptoms including visual disturbance and optic nerve damage [24]. That is a poor trade for no performance advantage.

2.6 Analyze Survivors

2.6.1 N_2O / IPA

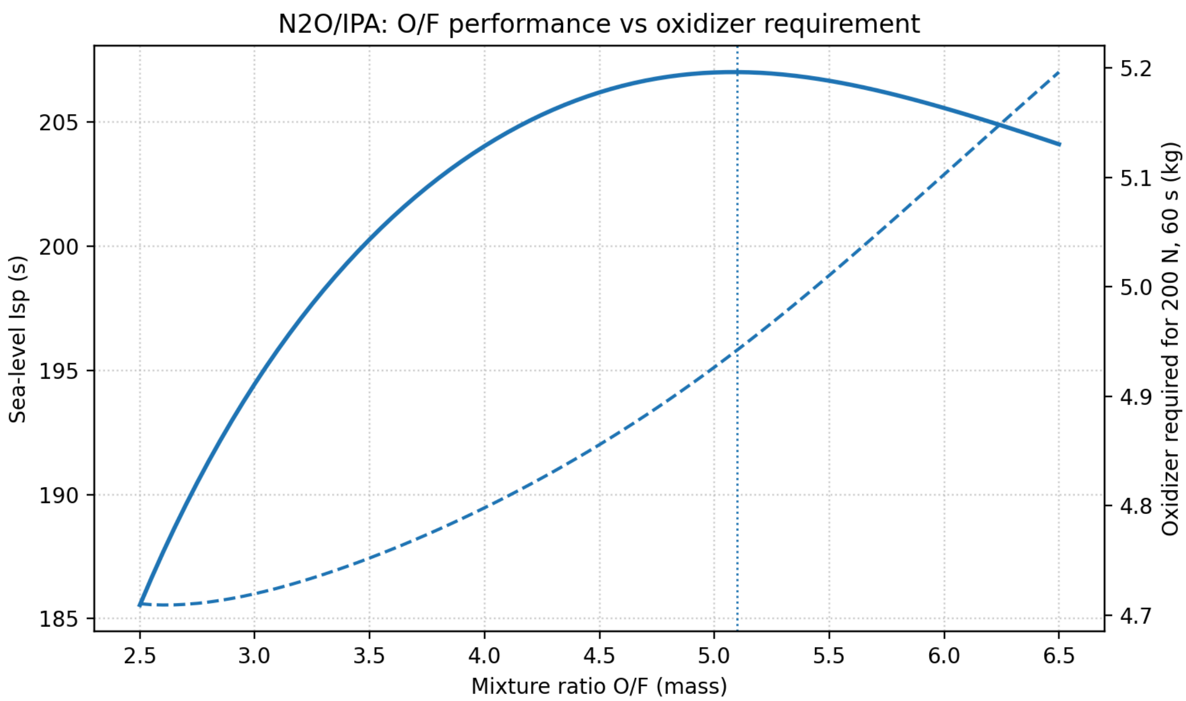

Figure 3: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

The O/F curve also shows why "run fuel-rich to save oxidizer" is not a large win for this pair. At the 200 N, 60 s reference burn, moving from O/F = 5.10 to O/F = 4.50 cuts N_2O from 4.94 kg to 4.86 kg, but loses 0.8 s of computed sea-level I_{sp} and adds about 0.11 kg of IPA. At O/F = 4.00, the oxidizer savings are only about 0.14 kg while the penalty grows to about 3.0 s. I would not move far below the optimum only for oxidizer mass; I would shift O/F only if injector stability, cooling, or hot-fire data asks for it [1].

N_2O/IPA survives because it best matches the actual hopper mission. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 207.0$s at O/F = 5.10 and $\varepsilon = 2.37$ [1]. That is not high compared with vacuum-optimized numbers, but it is enough for a minimal hopper and avoids the hardware burden of high-pressure GOX storage.

The most important external evidence is direct VTVL heritage. Gruyère Space Program's Colibri vehicle uses an N_2O/IPA bipropellant engine and is documented as a 2.45 m VTVL demonstrator with up to 1.25 kN thrust [25]. European Spaceflight reported the October 18, 2024 free flight to 105 m altitude, 30 m lateral translation, and 60 s duration [26]. That is the closest public analog to this project.

Decision **Selected baseline.** The main risk is not performance; it is injector/feed calibration for flashing N_2O. The advantage is compact oxidizer storage, no oxidizer regulator, cheap fuel, and direct small-VTVL precedent.

2.6.2 N_2O / Ethanol

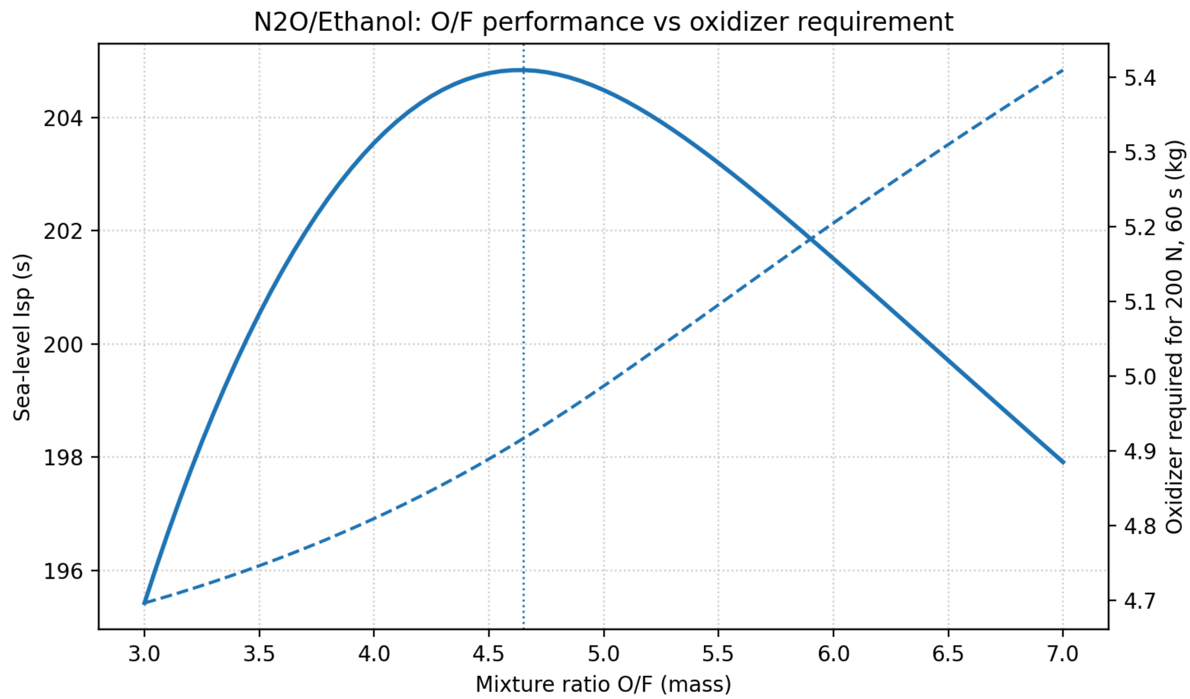

Figure 4: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

For N_2O/ethanol the curve is broad near the optimum. O/F = 4.20 reduces N_2O from 4.92 kg to 4.84 kg and costs only 0.6 s, so a small fuel-rich bias is acceptable if it helps cooling or calibration. Going to O/F = 4.00 saves only about 0.11 kg oxidizer and already loses 1.3 s; O/F = 3.50 saves about 0.17 kg but loses 4.3 s. The mass benefit is too small to justify a large performance departure by itself [1].

N_2O/ethanol is the strongest backup because it has broader academic static-fire literature and better coolant thermal conductivity. The sea-level ambient-corrected computed performance is $I_{\mathrm{sp,SL}} = 204.8\mathrm{s}$ at O/F = 4.65 and $\varepsilon = 2.38$ [1]. The difference from N_2O/IPA is only 2.2 s in the baseline comparison, so it should not drive the decision.

Ethanol has slightly lower specific heat but higher thermal conductivity than IPA at room temperature; the coolant trade is therefore not one-sided. IPA has slightly better bulk heat capacity, while ethanol has better film-side conduction margin [27].

Decision **Backup finalist.** Use ethanol if the design is intentionally aligned with N_2O/ethanol literature or if regen-cooling margin dominates fuel procurement convenience.

2.6.3 N_2O Architecture Risk: Feed, Injector, and Environment

This is the most important risk record in the propellant decision, because it describes the one problem the N_2O finalists cannot close on paper. N_2O is stored as a saturated liquid at its own vapor pressure. Resonac lists high-purity N_2O vapor pressure as 5.24 MPa at 20°C and gives a critical temperature of 36.41°C with critical pressure 7.24 MPa [28]. This enables self-pressurization, but it also means feed pressure changes strongly with tank temperature. It also changes during the burn: as liquid leaves, the remaining propellant evaporates to refill the ullage, the tank self-cools, and feed pressure sags, so thrust decays at a fixed valve position [29]. A 60 s hover is exactly the burn where that matters.

▷ **Las Vegas design-basis implication**

A N_2O tank in direct summer sun can approach or exceed the 36.4°C critical temperature. The baseline control map should be built around measured tank temperature and pressure, and the pressure-vessel MEOP check should use the highest credible tank temperature, not just nominal room temperature. Testing in shade or morning conditions is not a convenience; it is part of keeping the feed model inside the calibrated regime.

When saturated liquid N_2O accelerates through valves and injector passages, pressure reduction can initiate flashing before the chamber. The standard single-phase incompressible equation is therefore only a bounding model:

$$\dot{m} = C_d A \sqrt{2\rho \Delta P}$$

N_2O injector design should bracket the flow with SPI and HEM and then use a non-equilibrium model such as Dyer/NHNE for the design estimate. Published work on self-pressurizing propellant flow and N_2O injectors supports this SPI/HEM/NHNE framing [30], [31], [32], [33].

Model	Assumption	Expected bias	Use in this project
SPI	Single-phase incompressible liquid.	Often overpredicts flashing N_2O flow.	Upper-bound geometry check only.
HEM	Two-phase thermodynamic equilibrium.	Often lower-bound for short injector residence times.	Lower-bound flow check.
Dyer / NHNE	Non-homogeneous, non-equilibrium interpolation between SPI and HEM using residence/bubble-growth times.	Best practical pre-test estimate for short rocket-injector passages.	Design-reference model, then calibrate by hot fire.
Empirical map	Valve command, tank state, and chamber response fitted from tests.	Most reliable inside tested range.	Required for throttle control.

The AEL Snark work is the most relevant single data point because it is a 300 N N_2O/IPA thruster built for a VTVL vehicle: this thrust class, this propellant pair, this mission [22]. Two of its measured results show how far the system sits from textbook hydraulics. First, the N_2O injector pressure drop was found to be almost independent of massflow, because flash-boiling in the control valve and flash-boiling in the injector elements balance each other; the familiar square-law relation between orifice pressure drop and massflow simply does not hold. Second, chamber pressure varied almost linearly with the throttle-valve command, and the closed-loop controller was designed around a second-order transfer function identified from test data, not derived from first principles [22]. At O/F near 5 the oxidizer is more than 80% of the total flow, so the side of the engine that resists analysis is also the side that dominates it.

The practical consequence is concrete. I cannot pre-compute a valve schedule, an injector pressure-drop margin, or a throttle map for the N_2O engine and trust them. I can only bracket them with SPI/HEM/NHNE and then buy the real answer with a calibration matrix of hot fires across tank temperatures and throttle setpoints. That campaign is a real budget line — about $670–$1,200 of propellant for ten 200 N, 60 s fires at the sourced prices — and it is the main technical reason not to over-score N_2O just because its tanks are compact.

N_2O also carries a hazard GOX does not: it is a monopropellant-capable oxidizer whose ullage can decompose energetically if an ignition source reaches it, and hydrocarbon contamination catalyzes that decomposition [21]. The mitigation overlaps with GOX practice — oxidizer-clean plumbing, no oil or grease, care with adiabatic compression — so choosing N_2O does not buy exemption from cleanliness discipline. The fair counterpoint from the same work is that N_2O decomposition kinetics are roughly six orders of magnitude slower than H_2O_2 at comparable conditions, which is part of why it remains the standard amateur oxidizer [21].

2.6.4 GOX / Ethanol

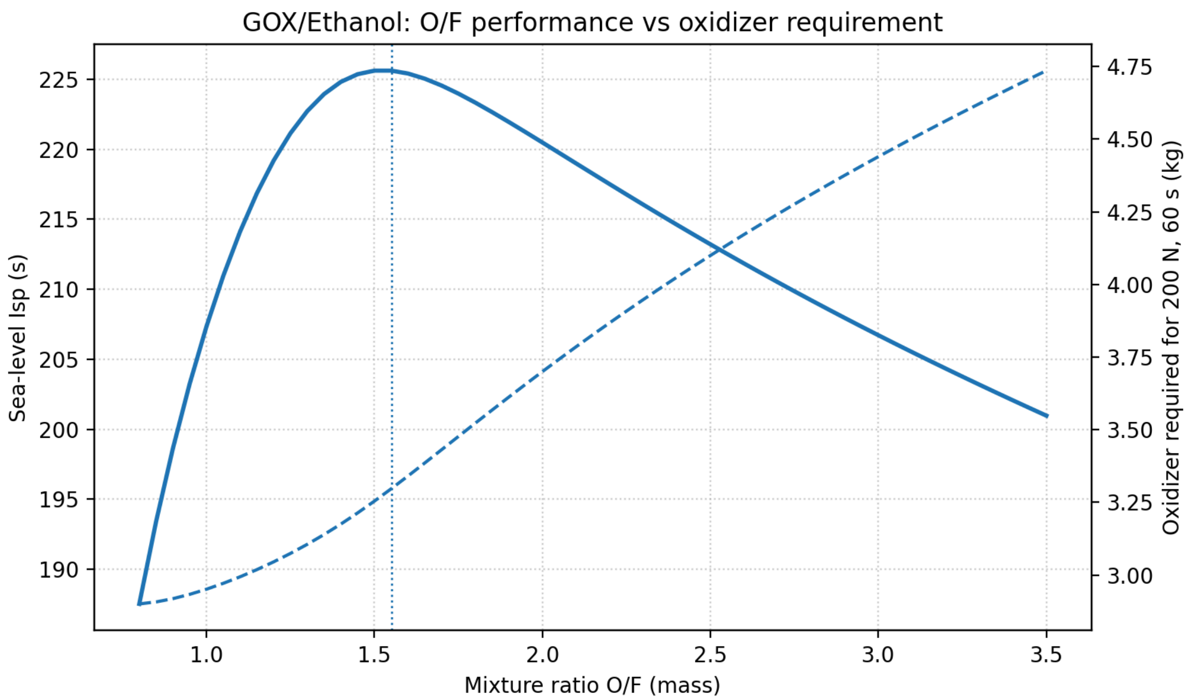

Figure 5: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

GOX/ethanol has a more useful fuel-rich knob because oxygen storage is the GOX penalty. At the 200 N, 60 s reference burn, O/F = 1.30 saves about 0.19 kg GOX, or 5.8%, with a 2.9 s computed I_{sp} loss; this may be worth checking if tank volume is the limiting item. Pushing to O/F = 1.20 saves 0.25 kg GOX, or 7.6%, but loses 6.4 s and increases ethanol mass. The useful trade window is narrow, around O/F = 1.30–1.55, not simply “as fuel-rich as possible” [1].

GOX/ethanol survives as the best alternative if the N_2O two-phase injector problem becomes unacceptable. The sea-level ambient-corrected computed performance is $I_{sp,SL} = 225.6s$ at O/F = 1.55 and $\varepsilon = 2.45$ [1]. It gives simpler single-phase oxidizer metering and stable regulator-fed pressure; the system-level cost of that simplicity is quantified in Section 2.4.

A NASA ignition characterization program tested GOX/ethanol at 150 psia and O/F = 1.8, making this the best-documented GOX/alcohol reference point near the baseline chamber pressure [34]. Oxygen-side hardware must be oxygen-cleaned and kept free of hydrocarbon contamination; the oxygen-cleaning and material-control burden is real, not paperwork [20].

Decision **Engineering-risk alternate.** Choose GOX/ethanol only if avoiding N_2O two-phase modeling is worth the larger gas storage, regulator cost, and oxygen-clean hardware process.

2.6.5 GOX / IPA

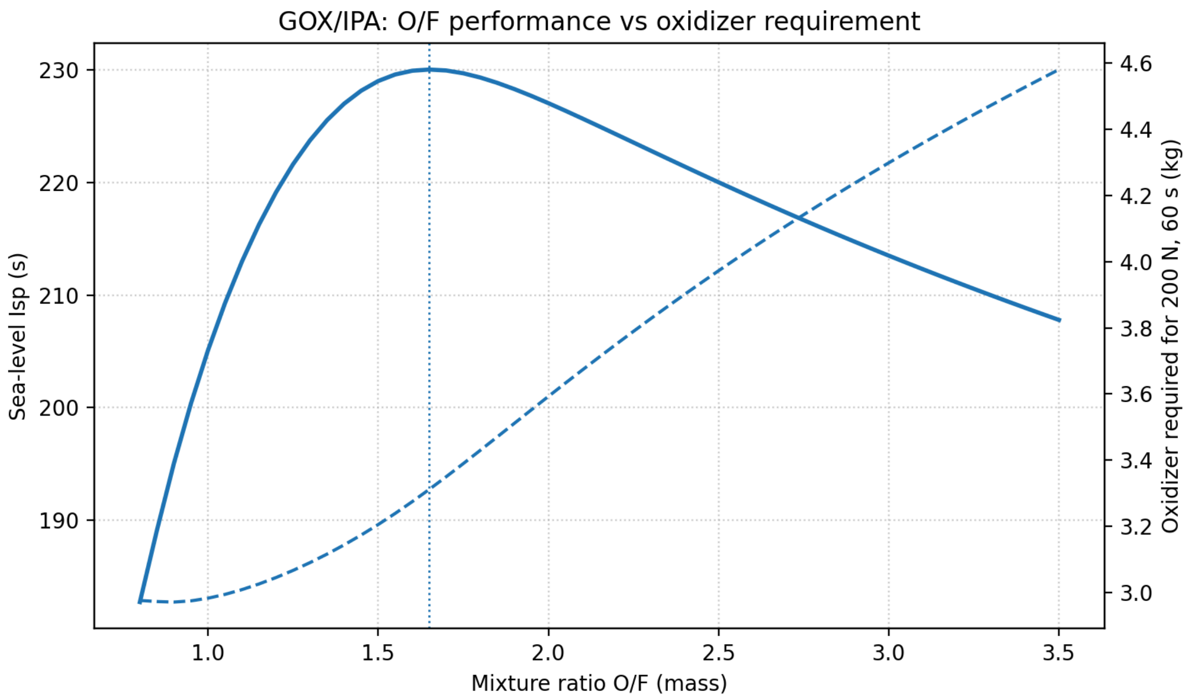

Figure 6: Individual O/F curve. The dashed curve in each panel is oxidizer mass required for a 200 N, 60 s burn.

GOX/IPA is less forgiving on the rich side. Dropping O/F from 1.65 to 1.40 saves about 0.17 kg GOX, or 5.1%, while losing 3.0 s, which could be tolerable if cylinder volume is marginal. Dropping to O/F = 1.20 saves only another 0.10 kg GOX beyond that but loses 10.9 s total, so the storage saving is not enough to justify the performance loss. If GOX/IPA is used, I would stay near O/F = 1.40–1.65 unless hot-fire stability points elsewhere [1].

GOX/IPA has the best sea-level ambient-corrected computed performance in this set: $I_{sp,SL} = 230.1$s at O/F = 1.65 and $\varepsilon = 2.43$ [1]. The fuel is cheaper than anhydrous ethanol, and per Section 2.4 that makes GOX/IPA the cheapest campaign in the whole set on consumables. The expensive pieces are the high-flow oxygen regulator, oxygen-compatible valves, and cleaning, not the cylinder; the itemized architecture cost is in Section 2.4.

Decision **Valid but not baseline.** GOX/IPA is a credible GOX alternate. It is not selected over GOX/ethanol because the main GOX advantage is single-phase oxygen metering, and GOX/ethanol has stronger nearby ignition-validation evidence.

2.7 Pump Option Check

The hard rule is no turbopumps, not no pumps. I am still keeping the first-article survivor comparison pressure-fed because the pump does not remove the dominant oxidizer problems. For N_2O, self-pressurization is the benefit, and published injector work still treats pump/supercharge conditions as ways to set tank state or avoid cavitation, not as a way to eliminate flashing physics [31]. For GOX, a small electric liquid pump is not available because cryogenic LOX is screened out, and a gas compressor/feed pump would add oxygen-clean rotating hardware, battery power, controls, and heat without solving the large gas-storage requirement.

A small electric fuel pump remains allowed. It may become attractive later if the fuel tank or fuel pressurant mass dominates the vehicle, especially because IPA/ethanol are ordinary liquids compared with the oxidizers. I am not adding it to the baseline yet because it adds a controller, battery/current margin, bypass or relief path, transient calibration, and a new single-point failure while leaving the oxidizer architecture unchanged.

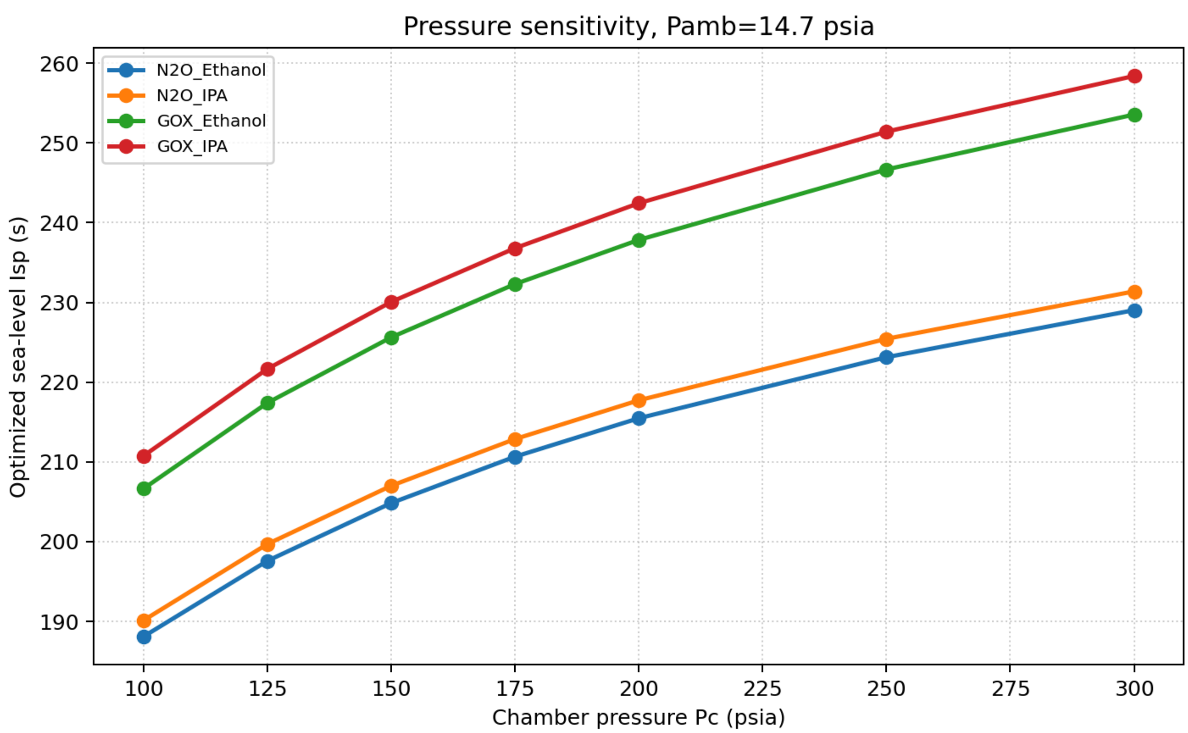

Figure 7: Pressure sensitivity for the four surviving alcohol finalists [1].

2.8 Cryogenic Reconsideration: LOX

The no-cryogenics rule was written from fear of complexity, so before selecting I am testing it against numbers instead of instinct — especially since both surviving oxidizers carry a structural penalty: GOX pays in tank tare, regulator hardware, and storage volume; N_2O pays in two-phase calibration and thermal sensitivity. LOX removes both at once. It is a dense, single-phase liquid at the injector (about 1,141 kg/m^3 at its normal boiling point [35]), it stores at low dewar pressure instead of as 2265 psia gas, and it is the cheapest oxidizer I can document: FAR sells 230 L, roughly 262 kg, for \$540 — about \$2/kg — and a trade source reports \$200-class quotes for a 180 L dewar [2], [5].

It also has the deepest small-VTVL heritage of any candidate in this notebook, which the hard screen ignored. Masten's Xombie flew a regeneratively cooled LOX/IPA engine 227 times, Xodiac flew pressure-fed LOX/IPA, and Armadillo's Pixel and Mod vehicles flew pressure-fed LOX/ethanol through the Lunar Lander Challenge

[36], [37], [38]. The propellant family with the most small hover-flight history is not N_2O/alcohol; it is LOX/alcohol.

The bill on the other side is operational, and it is real: cryo-rated valves and seals, thermal contraction and icing, a chilldown procedure, boiloff that forbids long holds and forces load-late operations, dewar logistics (deposit, rental, transport), the same CGA G-4.1 oxidizer cleanliness as GOX [20], and oxygen-enrichment PPE. Decisive at this scale: LOX is not usefully self-pressurizing, so a pressure-fed LOX system reintroduces the high-pressure pressurant bottle and regulator that the N_2O architecture avoids. It buys back GOX-style feed hardware and adds cryogenic operations on top of it.

The screen therefore stands for the first article, but as a recorded decision with a reopen condition rather than a closed door: if the N_2O calibration campaign runs past roughly 15–20 hot fires without converging, or the field thermal envelope cannot be held, LOX/IPA is the rational pivot with the most direct heritage, and the no-cryogenics requirement would be formally waived at that point.

2.9 Select Best Solution

The high GOX performance and simple single-phase metering are real, and the campaign figure in Section 2.4 shows GOX/IPA is genuinely the cheapest pure static-fire program, paying back its hardware premium in about 15 hot fires. But the requirement is a hopper, and the quantified GOX penalty is really two distinct penalties. The budget penalty is $1,190–$1,870 of oxidizer-side hardware, nearly all of it regulator and oxygen-clean components rather than the cylinder. The flight penalty is the decisive one: mapped across the whole thrust/burn envelope, GOX storage carries about 7 kg of steel per kilogram of usable oxygen against about 1.5 kg/kg for a nitrous bottle, and a flight-weight composite oxygen vessel would fix the tare while making the budget worse. The high peroxide density is also real, but sourcing and decomposition risk dominate.

N_2O wins as an architecture, not as a propellant. Per kilogram it is the most expensive oxidizer in the finalist set, and Section 2.6.3 records that it is the only finalist whose injector and throttle cannot be closed analytically, so the selection explicitly buys a calibration campaign. What it buys back is that the storage bottle is also the pressurization system, the first fire is the cheapest to reach, and the closest public analog to this exact mission flew on it [25].

> **Decision** **Baseline propellant selection: N_2O / IPA.** N_2O/IPA is not selected because it is analytically easy; it is selected because it best fits the full hopper constraint set: compact self-pressurizing oxidizer storage, cheap and accessible fuel, and the most relevant small-VTVL flight precedent. The design must explicitly budget the N_2O two-phase modeling and hot-fire calibration matrix as its own line item — about $670–$1,200 of propellant for a ten-fire matrix at 200 N, 60 s.

The best alternatives are conditional. **GOX/ethanol** is the cleanest technical alternate if avoiding two-phase N_2O behavior is worth the tank/regulator mass and cost. **N_2O/ethanol** remains the fuel backup if the project chooses to align with N_2O/ethanol literature or needs ethanol's higher thermal conductivity for cooling margin. **GOX/IPA** is credible and has the best computed I_{sp}, but it is not the reference GOX fuel because the fuel choice does not solve the GOX storage/regulator problem and GOX/ethanol has stronger nearby ignition-validation evidence. **LOX/IPA** is the documented growth path: Section 2.8 records the reopen condition and the flight heritage that supports it.

3 Engine Sizing

4 Nozzle Design

5 Injector Design

6 Feed System Design

7 Cooling Analysis

8 Structural Analysis

9 CAD Design

10 Simulation

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